

1. What is the average GRE score of an applicant who has been accepted into a PhD program?

I ran a split on this data because it just wasn't jiving with what I was looking at. Here is what I found:

All PhD GRE Scores: Average of 264.72, quantity 754

Combined GRE Score Range (260-340): Average of 325.9, quantity 463

Single GRE-A Score Range (130-170): Average of 166.69, quantity 290

Meaning this is definitely mixed data (with one more additional outlier that didn't get caught in the cleaning since $463+290 \neq 754$). While a literal average across all valid entries yields 264.72, this figure is not meaningful due to the mixed scales in the data. The more defensible answer is 325.90, based on the subset of entries falling into the combined GRE score range, which represents the majority of the internally-consistent data.

2. What is the average GPA of an applicant who has been accepted into a Master's program?

Due to the bounds checking we implemented I was more confident with the results of this, average of 3.706 with 6,988 applicants.

3. What are 2-3 shortcomings of inferring decision_date from a text status field and the term value?

The date associated with the status change could be auto-generated by GradCafe (I don't know the source website well enough) when the user clicks a new status, which could give a date significantly varied from the actual status change.

Doing a simple year move could end up wrong if the acceptance was for, say, Spring 2026 and the status change occurred in December 2025. An additional check for this should be performed.

Assuming there is no text validation, there could be bad data in the year, causing an error on conversion (though the consistency that I saw in the data makes me think this is probably not the case).

4. What did your correlation analysis suggest about the relationship between GPA and GRE-related measures?

GRE-GRE V Coefficient: 0.0049, p-value: 0.7943

GRE-GPA Coefficient: -0.0449, p-value: 0.0122.

Based purely on the coefficient, the GRE and GRE-V has little correlation to each other (being close to 0). However, the high p-value would indicate that there is a high probability that this is an effect of sampling error. The relation between GPA and GRE seems to have a slight negative correlation, but it is still rather small (also being close to 0). The incredibly low p-value indicates that this sampling was likely to be an accurate representation of the correlation between them. Due to the issue with the GRE data, this might not actually be reflective of anything and dividing the scores may get us very different results.

5. Did your group comparison suggest that accepted and rejected applicants differ in GPA? Briefly explain.

Test used: Mann-Whitney U

U statistic: 68,629,801.00

p-value: 5.727e-05

Accepted group: 11,658 entries, median GPA=3.810

Rejected group: 12,139 entries, median GPA=3.830

The Mann-Whitney U test seemed more appropriate, since the “ttest_ind” assumes normality (grades for grad students will be clustered toward the upper end of the 0-4.0 scale). The incredibly low p-value in this case would suggest that there is a statistical significance in the relation of GPA to Acceptance Status. Looking at the medians it would actually appear that the Rejected group tend to have slightly higher GPAs. This suggests two additional possibilities, that acceptance is not purely related to GPA and/or that the sampling size we have for this comparison is not large enough to gain a truly accurate picture.

6. What did your chi-square test suggest about the relationship between your selected categorical variables?

Contingency Table:

	Accepted	Rejected	Waitlisted
Master's	9471	3175	883
PhD	7916	18327	3445
PsyD	206	169	57

Chi-square statistic: 7330.5549

p-value: 0

Degrees of freedom: 4

Based on the large chi-square statistic and 0 (or basically 0, if the number was just too small for the float), there is strong evidence to indicate that degree type and acceptance rate are not independent. Honestly though, looking at the contingency table makes this pretty clear, since PhD can be seen to reject more than twice the number of acceptances, while the Master's has nearly three times as many acceptances per rejection. What this does not tell us, is the why of this. It does give us a starting point to do some more digging though.

7. What data quality issues still remain in the cleaned dataset, even after your processing?

- I did note that the GRE field itself is problematic, and the plot comparing GRE and GRE-V actually bears this out, showing two clearly divided groupings around the combined GRE-A and GRE-V score (one way of interpreting just 'GRE') and just the value for the GRE-A (the other way of interpreting GRE). This also throws the fit-line in the chart off. While we can't know how the combined score version divides out, we could make a reasonable assumption that entries in a certain range are the combined score and we could create new columns to divide these two.
- The LLM pass on the Degree and University were pretty good, but could possibly be better.
- The year assumption based on term start date could be wildly wrong and there is no real way to check this.
- Categories for early, mid, late cycle are best guess, but still arbitrary, because each university sets their own deadlines and term start dates.
- And, inherent in any dataset generated from self-reported data... it is self-reported so may not be reflective of reality or could only be a certain subset of applicants that have natural tendencies that could go counter to the rest of the population. Not that there is much that can be done about this particular item other than switching sources (maybe to info from the Universities themselves?).